

LED Drivers Driving New Lighting Trends

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Compact LED driver by GlacialTech

As the lighting industry moves towards energy-efficient solutions like light-emitting diodes (LEDs), many key players are launching LED products for the Indian market.

According to Frost & Sullivan, the overall Indian LED lighting market is expected to reach \$399.2 million by 2015, at an annual growth rate of 41.5 per cent till 2014. This, in turn, has given a boost to the market for components like LED drivers, which have a tremendous potential to grow in various sectors such as general lighting, automotive lighting and industrial lighting.

LED driver is an essential component of LED lighting, which controls the amount of current and voltage supplied to the LEDs. With an increasing market size, LED driver technology is also getting more advanced. Most LED drivers are used in backlights for LCD displays, such as in mobile phones and TVs. LED drivers are also used in DVD and MP3 players.

What's the latest?

With LED lights getting more advanced with respect to technology, there is a need for maximum integration with LED drivers to make LED lights more cost-effective and compact in size. Customers look for features like power factor correction (PFC), triac and dimming.

There are many different types of LED drivers—white LED drivers, LED PWM drivers, RGB LED drivers, constant-current LED drivers and 7-segment LED drivers. White LED drivers provide white light for backlighting with very low noise and efficiencies as high as 90 per cent. LED PWM drivers can be programmed through an I²C compatible interface for applications that require colour, pattern and intensity programmability control. In RGB LED drivers, all the functions are controlled by software through an SPI interface and internal registers. Constant-current LED drivers feature internal circuitry that monitors the loop current of each LED and

automatically adjusts the DC voltage generated to the minimum value required to produce the highest forward voltage. 7-segment LED drivers use a combinational logic circuit, which accepts a 4-bit BCD input and generates seven output signals to control seven individual display segments.

Dimming capability. Today, LED drivers are available that can dim the LED light output from 100 per cent to zero. During dimming, LEDs still operate at the same voltage and current as during full light output. In addition, lamp life is not affected by dimming, as is sometimes the case with frequently dimmed fluorescent lighting.

These LED drivers come in power ratings ranging from 2W to 150W. Highly efficient DC-input LED drivers in buck and buck-boost topology are also available.

ON Semiconductor has introduced two new LED drivers to help designers optimise their solutions for automotive, industrial and general lighting applications. The new power-factor-corrected offline LED driver (NCL30051), when combined with NCL30160 dimmable constant-current step-down LED driver, provides a good solution for high-brightness applications where space is limited and high efficiency important.

Buck-boost LED drivers. RECOM Asia has launched RBD LED drivers, which are highly efficient, 3-20W DC-DC constant-current LED drivers in buck-boost topology. These LED drivers allow the users to drive high-power LEDs even from a battery source, making them a perfect solution for solar applications.

The RECOM RBD 12 series, which the company claims to be the only buck-boost module solution in the